

COMPLETE-SHELL COLLISION BOUNDS FOR THE ERDŐS–STRAUS EQUATION: A POINTWISE OBSTRUCTION AND RETAINED NONPRINCIPAL CERTIFICATES

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ABSTRACT. For the complete marked first-half divisor-pair source, we prove that isolating only the principal character and bounding the rest with the adverse sign gives a summed lower estimate at most $-p \log p/24$ for every prime $p \equiv 1 \pmod{4}$, $p \geq 2^{20}$. The same bound holds when at most J even-character modes are retained per shell and $p \geq 2^{20}J^3$, including adaptive subgroup observations. We retain nonprincipal even information through unsaturated cosets and give a sharp integer balancing bound, a complement-aware refinement, free fourfold original pair orbits, and complete marked denominator returns. All four failures of the principal-only test in the stated two-million hard-prime census have certified nonprincipal or integer repairs. A proper five-mode observation also attains the target lower bound at a prime-order quotient with no intermediate subgroup. A universal positive lower bound for the full ES pair sum is not established.

1. OBJECTIVE AND THE PRECISE OUTCOME

The Turn 5 handoff asks for positivity of the complete original first-half pair sum at a prescribed prime, not for another family constructed after useful factors are supplied. We make that attempt here. The exact pair source and its $2\tau_{\text{div}}(h)$ fibres are inherited from [2, §§3–4]; the Type I/II coordinates are classical [1, §2]. They are rederived below so that the new estimates have a complete arithmetic return.

The principal-character isolation attempt fails quantitatively: for every prime $p \equiv 1 \pmod{4}$ with $p \geq 2^{20}$, its complete unweighted lower estimate is at most $-p \log p/24$. More generally this remains true when at most J even characters are retained at each shell, once $p \geq 2^{20}J^3$, including arbitrary shellwise subgroup observations of index at most J . An all-exponent refinement extends this obstruction to every fixed sublinear mode budget, with an explicit onset. This is an obstruction to that specified estimate, not to ES or to every possible signed estimate. An original finite prime also defeats *every* nonnegative shell reweighting of the principal-only estimate.

The surviving calculation retains nonprincipal even-character information through unsaturated cosets and uses an exact integer collision bound. It returns actual states at the four bounded principal-only failures, including a cubic quotient whose two occupied coarse coefficients are both eight. No theorem that these improved bounds are positive at some shell of every prime is proved. No historical-priority claim is made for the estimates or the classical Cauchy–Schwarz, balancing, Fourier, and divisor arguments they use.

2. COMPLETE SOURCE, FIRST-HALF COMPLETENESS, AND ORIGINAL RETURNS

Let $p \equiv 1 \pmod{4}$ be prime and put

$$I_p = \{a \in \mathbb{Z} : p/4 < a < p/2\}, \quad R_a = 4a - p.$$

Thus $R_a = 3, 7, \dots, p-2$, and $\gcd(pa, R_a) = 1$. Write $a = \prod_{\ell} \ell^{e_{\ell}}$. The complete original divisor source is

$$\beta_{\ell} \in [-e_{\ell}, e_{\ell}]_{\mathbb{Z}}, \quad u = \prod_{\ell} \ell^{e_{\ell} + \beta_{\ell}} \mid a^2. \quad (1)$$

Its inverse is $\beta_{\ell} = v_{\ell}(u) - e_{\ell}$. The gates and normalization are

$$E : R_a \mid 4u + 1, \quad M : R_a \mid 4u + p,$$

$$d_0 = \gcd(a, u), \quad (h, r, s) = (d_0^2/u, u/d_0, a/d_0), \quad a = hrs, \quad u = hr^2, \quad \gcd(r, s) = 1.$$

Prime valuations prove the asserted positive integers. The ordered returns are

$$E : \kappa = (pr + s)/R_a, \quad (x, y, z) = (a, hsk, phrk), \quad (2)$$

$$M : \lambda = (r + s)/R_a, \quad (x, y, z) = (a, phs\lambda, phr\lambda). \quad (3)$$

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The gates and the fact that h, r, s are units modulo R_a prove the integer quotients. Clearing denominators proves $4/p = 1/x + 1/y + 1/z$. The ordered divisor inverses are respectively

$$u = \frac{a^2}{R_a y - pa}, \quad u = \frac{pa^2}{R_a y - pa}. \quad (4)$$

The middle involution $u \mapsto a^2/u$ swaps the last two denominators; exterior orientation retains its unique p -divisible denominator last.

For completeness, every positive solution has a distinguished first-half coordinate. Order $x \leq y \leq z$. Then $p/4 < x \leq 3p/4 < p$, and at least one, but not all, denominators are multiples of p . In Type II, $y, z \geq p$ force $x < p/2$. In Type I write the two non- p denominators as dr, ds , with $r \leq s$ and $\gcd(r, s) = 1$, and the third as pk . The equation gives $p \mid 4k - 1$; put $t = (4k - 1)/p \equiv 3 \pmod{4}$. Then

$$tdrs = k(r + s), \quad k = hrs, \quad r + s = t\lambda, \quad d = h\lambda.$$

Here $rs \mid k$ and $\gcd(t, h) = 1$ justify the two divisions. The case $r = s$ would give $t \mid 2$ and is impossible. For $r < s$,

$$p - 2hr\lambda = \frac{2hr(s - r) - 1}{t} > 0.$$

So $hr\lambda$ is a first-half denominator. Finally

$$(R_a y - pa)(R_a z - pa) = p^2 a^2 \quad (5)$$

classifies a solution at this denominator. Both factors are positive; their p -valuations are $(0, 2), (1, 1), (2, 0)$. The required congruence modulo R_a and the divisor relation recover exactly (1)–(4), with the recorded exterior orientation or either middle order. These are the inherited Type I/II coordinates [1, Propositions 2.2 and 2.6], not a new parametrization.

2.1. The common pair source. Retain actual divisors of pa as $d = p^\epsilon v$, where $v \mid a$ and $\epsilon \in \{0, 1\}$. Put

$$\mathcal{P}_a = \{(b, c) : b, c \mid pa, R_a \mid b + c\}, \quad T_a = |\mathcal{P}_a|. \quad (6)$$

A primitive labelled M pair (r, s) has the full fibre

$$(b, c) = (p^\epsilon dr, p^\epsilon ds), \quad d \mid h, \quad \epsilon \in \{0, 1\}; \quad (7)$$

an E pair has fibre

$$(pdr, ds) \text{ or } (ds, pdr), \quad d \mid h. \quad (8)$$

To invert, divide (b, c) by its gcd. If neither coprime part contains p , use M and retain the gcd's p -bit. Otherwise remove its one p , designate that side r , and use E, retaining the side. The lcm condition gives $drs \mid a$ and $h = a/(rs)$. Thus every original state has exactly $2\tau_{\text{div}}(h)$ preimages. At fixed a the map $(r, s, h) \mapsto u = hr^2$ is the original divisor inverse. Consequently, for $F_a = |E_a| + |M_a|$,

$$2F_a \leq T_a \leq 2\tau_{\text{div}}(a)F_a, \quad \text{ES}(p) \iff \sum_{a \in I_p} T_a > 0. \quad (9)$$

The full exact-state Möbius inversion remains in [2, §4, “Complete primitive count”]; it is not an additional prerequisite for using (9). Here τ_{div} is the divisor-count function, not the absent-support element τ .

Lemma 2.1 (Original fourfold orbits). *For every actual shell, $T_a \in 4\mathbb{Z}_{\geq 0}$. The involutions*

$$S(b, c) = (c, b), \quad I(b, c) = (pa/b, pa/c) \quad (10)$$

generate a free Klein four action on $\mathcal{P}_a$.

Proof. They commute and preserve (6). A fixed point of S would give $R_a \mid 2b$, contrary to coprimality and $R_a \geq 3$. A fixed point of I would give $b^2 = pa$, contrary to $v_p(pa) = 1$. A fixed point of SI gives $bc = pa$ and $b \equiv -c$. Since $pa \equiv 4a^2 \pmod{R_a}$, it would make -1 a square modulo R_a . This is impossible when $R_a \equiv 3 \pmod{4}$: a prime $3 \pmod{4}$ occurs to odd exponent in R_a and -1 is a nonsquare there. Thus all orbits have four members. In (7), S swaps r, s and I additionally sends (d, ϵ) to $(h/d, 1 - \epsilon)$. In (8), S changes the side and I also replaces d by h/d . These are actions on the original marked fibres, not extra ES states. $\square$

3. THE FULL-SHELL PRINCIPAL ESTIMATE IS UNIFORMLY NEGATIVE

In $G_a = (\mathbb{Z}/R_a\mathbb{Z})^\times$, let

$$c_a(g) = \#\{d \mid pa : d \equiv g \pmod{R_a}\}, \quad m_a = 2\tau_{\text{div}}(a), \quad \mathcal{E}_a = \sum_g c_a(g)^2.$$

Character orthogonality gives

$$T_a = \frac{1}{\varphi(R_a)} \sum_{\chi} \chi(-1) \left| \sum_{d \mid pa} \chi(d) \right|^2, \quad \mathcal{E}_a = \frac{1}{\varphi(R_a)} \sum_{\chi} \left| \sum_{d \mid pa} \chi(d) \right|^2. \quad (11)$$

All characters, including imprimitive ones, are retained. The first identity is already present in the historical pair-source transcript [3]; its use here does not claim a new Fourier reformulation. The principal contribution is $P_a = m_a^2/\varphi(R_a)$. Giving every nonprincipal term the adverse sign yields

$$T_a \geq L_a := 2P_a - \mathcal{E}_a = \frac{8\tau_{\text{div}}(a)^2}{\varphi(R_a)} - \mathcal{E}_a. \quad (12)$$

This is a valid lower bound, but the next theorem shows why summing it cannot close the conjecture.

Lemma 3.1 (Two elementary explicit bounds). *For $n \geq 1$, $\tau_{\text{div}}(n) < 4n^{1/3}$. For $x \geq 3$,*

$$\sum_{\substack{r \leq x \\ r \equiv 3 \pmod{4}}} \frac{1}{\varphi(r)} < \frac{7}{5} \left(1 + \frac{1}{4} \log x \right). \quad (13)$$

Proof. For primes at least eleven, $e+1 \leq 2^e \leq q^{e/3}$. For $q = 2, 3, 5, 7$, successive ratios show that $(e+1)/q^{e/3}$ is maximized at $e = 3, 2, 1, 1$, respectively. The four maxima are bounded by $2, 3/2, 6/5, 11/10$; their product is $3.96 < 4$. The ratio check after a maximum is $(e+2)^3 < q(e+1)^3$ and becomes stronger as e increases; before the maximum the reverse inequalities hold. This proves all exponent cases, not only the finitely evaluated maxima.

For the second assertion use the classical divisor identity, also recorded in [4, Lemmas 8–10, pp. 11–12]. Only this identity and its elementary expansion are used; no probability or independence assertion from that paper enters the present proof. Explicitly,

$$\frac{r}{\varphi(r)} = \sum_{d \mid r} \frac{\mu(d)^2}{\varphi(d)}.$$

Writing $r = dm$, both variables are odd and m is in one specified reduced class modulo four. For either class, $\sum_{m \leq X} 1/m \leq 1 + \frac{1}{4} \log X$: isolate its first term and bound later terms by the preceding length-four integrals. Consequently the left side of (13) is at most

$$\left(1 + \frac{1}{4} \log x \right) \sum_{d \text{ odd}} \frac{\mu(d)^2}{d\varphi(d)} = \left(1 + \frac{1}{4} \log x \right) \prod_{q \text{ odd prime}} \left(1 + \frac{1}{q(q-1)} \right).$$

Absolute convergence justifies this expansion. Its product is at most

$$\exp \left(\sum_{j \geq 1} \frac{1}{2j(2j+1)} \right) = \exp(1 - \log 2) = \frac{e}{2} < \frac{7}{5}.$$

The last strict inequality follows from the factorial series for $e < 14/5$. □

Theorem 3.2 (A complete pointwise failure of principal domination). *For every prime $p \equiv 1 \pmod{4}$ with $p \geq 2^{20}$,*

$$\boxed{\sum_{a \in I_p} L_a \leq -\frac{1}{24} p \log p.} \quad (14)$$

Proof. Put $\mathcal{P} = \sum_a P_a$ and $\mathcal{D} = \sum_a \mathcal{E}_a$. Because $a < p/2$ and $2^{-2/3} < 2/3$, Lemma 3.1 gives

$$\mathcal{P} < \frac{224}{15} p^{2/3} (\log p + 4). \quad (15)$$

On the other hand, $\mathcal{E}_a \geq m_a = 2\tau_{\text{div}}(a)$ from the original diagonal pairs. Let $Y = \lfloor \sqrt{p}/2 \rfloor$. Every divisor $d \leq Y$ of a has a distinct complementary divisor $a/d > Y$, since $a > p/4 \geq Y^2$. The interval I_p contains more than $p/(4d) - 1$ multiples of d , so

$$\mathcal{D} \geq 2 \sum_a \tau_{\text{div}}(a) > p \sum_{d=1}^Y \frac{1}{d} - 4Y > p(\frac{1}{2} \log p - \log 2) - 2\sqrt{p}. \quad (16)$$

The harmonic bound is $\sum_{d \leq Y} 1/d \geq \log(Y+1)$, and $Y+1 > \sqrt{p}/2$.

For $p \geq 2^{20}$, one has $p^{1/3} > 100$, $\log p > 40/3$, $\log 2 < 3/4$ and $2/\sqrt{p} \leq 1/512$. Equations (15)–(16) imply

$$\frac{2\mathcal{P}}{p \log p} < \frac{448}{15} \frac{1}{100} \frac{13}{10} = \frac{728}{1875}, \quad \frac{\mathcal{D}}{p \log p} > \frac{7}{16}.$$

Finally $7/16 - 728/1875 = 1477/30000 > 1/24$. This proves (14). The elementary logarithm bounds follow either from integration or the rational series retained in the certificate. No distribution-of-primes theorem or ES occupancy enters. $\square$

Corollary 3.3 (Adaptive finite quotient resolution). *For each shell choose any subgroup $B_a \leq G_a$ containing -1 , with $[G_a : B_a] \leq J$. Let $m_{a,C} = \sum_{g \in C} c_a(g)$ and put*

$$L_{a,B_a} = \frac{2}{|B_a|} \sum_{C \in G_a/B_a} m_{a,C}^2 - \mathcal{E}_a. \quad (17)$$

For every integer $J \geq 1$ and prime $p \equiv 1 \pmod{4}$ with $p \geq 2^{20} J^3$,

$$\sum_a L_{a,B_a} \leq -\frac{1}{24} p \log p. \quad (18)$$

The subgroups may depend on the actual factorization of each shell. More generally, let Σ_a be any at most J even characters, with conjugate pairs retained when a real observation is intended, and put

$$L_{a,\Sigma_a} = \frac{2}{\varphi(R_a)} \sum_{\chi \in \Sigma_a} \left| \sum_{d|pa} \chi(d) \right|^2 - \mathcal{E}_a. \quad (18a)$$

Then $T_a \geq L_{a,\Sigma_a}$ and the same summed upper bound (18) holds.

Proof. Let $\mathcal{E}_+$ be the full even-character energy. Equation (11) gives $T_a = 2\mathcal{E}_+ - \mathcal{E}_a$. Retaining any part of that nonnegative energy proves (18a). Each squared character sum is at most m_a^2 , so $L_{a,\Sigma_a} \leq 2JP_a - \mathcal{E}_a$. The previous proof applies because $p^{1/3} > 100J$. The subgroup version is $\Sigma_a = B_a^\perp$, whose cardinality is its index. Equivalently use $\sum_C m_{a,C}^2 \leq m_a^2$ in (17). $\square$

The same argument permits positive weights $w_0 \leq w_a \leq w_1$ after replacing J by Jw_1/w_0 in the height condition, and yields the upper bound $-w_0 p \log p / 24$. It does not cover truncating all negative bounds to zero, weights without a positive lower bound, unbounded-index observations, or the integer refinement below. Those restrictions are part of the theorem.

3.1. An effective obstruction at every fixed sublinear spectral scale. The power J^3 above is not an intrinsic barrier. For an integer $m \geq 3$, put

$$K_m = \prod_{\substack{q < 2^m \\ q \text{ prime}}} \max_{0 \leq e < 2m} \frac{(e+1)^m}{q^e}, \quad C_m = \min\{C \in \mathbb{Z}_{>0} : C^m \geq K_m\}. \quad (18b)$$

These are finite, exactly computable constants. For every integer $a \geq 1$,

$$\tau_{\text{div}}(a) \leq C_m a^{1/m}.$$

For $q \geq 2^m$, use $e+1 \leq 2^e$. For a smaller prime, the consecutive ratio is decreasing, and once $e+1 \geq 2m$,

$$\left(1 + \frac{1}{e+1}\right)^m \leq \left(1 + \frac{1}{2m}\right)^m \leq \sum_{j=0}^m \frac{1}{2^j j!} < 2 \leq q.$$

The maximum is therefore among the stated $2m$ exponents. Multiplication proves the all-integer bound.

Theorem 3.4 (Effective sublinear-resolution obstruction). *Retain any at most J even characters at each original shell. For every prime $p \equiv 1 \pmod{4}$ such that*

$$p \geq 2^{20}, \quad p^{1-2/m} \geq 12JC_m^2,$$

$$\boxed{\sum_{a \in I_p} L_{a, \Sigma_a} \leq -\frac{1}{8}p \log p.} \quad (18c)$$

For $m = 4$, $C_4 = 9$, so it suffices that

$$\boxed{p \geq \max\{2^{20}, 944784J^2\}.} \quad (18d)$$

Proof. The reciprocal-totient estimate (13) gives

$$\mathcal{P} \leq \frac{7}{5}C_m^2 2^{-2/m} p^{2/m} (\log p + 4).$$

As before $\mathcal{D}/(p \log p) > 7/16$ and $1 + 4/\log p < 13/10$. The capacity inequality therefore gives

$$\frac{2J\mathcal{P}}{p \log p} < \frac{91}{300}, \quad \frac{7}{16} - \frac{91}{300} = \frac{161}{1200} > \frac{1}{8}.$$

This proves (18c). At $m = 4$, the primes 2, 3, 5, 7, 11, 13 have maximizing exponents 5, 3, 2, 1, 1, 1, and

$$K_4 = \frac{127401984}{25025}, \quad 8^4 < K_4 < 9^4.$$

Thus $C_4 = 9$, and $12C_4^2 = 972$, whose square is 944784. $\square$

For any fixed $\delta > 0$, choose m with $2/m < \delta$. The theorem then excludes a positive raw sum from any per-shell budget at most $p^{1-\delta}$ for every

$$p \geq \max\{2^{20}, (12C_m^2)^{1/(\delta-2/m)}\}.$$

This is a bound on the stated adverse-sign estimate, not a lower bound on the number of modes needed to locate an ES witness by any method. Zero truncation, integer balancing and arithmetic control of the omitted signs remain outside this obstruction.

The diagonal mass in (16) is exactly cancelled in the true signed formula, since $\sum_{\chi} \chi(-1) = 0$. In particular one may write (11) with each squared character sum replaced by $|\sum_{d|pa} \chi(d)|^2 - m_a$. That exact removal does not itself bound the remaining signed terms. The theorem rules out the specific adverse-sign replacement, not arithmetic cancellation in the original sum.

4. RETAINING NONPRINCIPAL INFORMATION ON UNSATURATED COSETS

Fix an actual shell and a chosen subgroup $B \leq G_a$ with $-1 \in B$; it is *not* assumed to be the actual stabilizer of the divisor support. Write $|B| = 2s$. Every coset is partitioned into s antipodal pairs $\{g, -g\}$. For integer $m \geq 0$, write $m = sq + r$ with $0 \leq r < s$ and define

$$\mathbf{b}_s(m) = sq^2 + r(2q + 1). \quad (19)$$

Theorem 4.1 (Integer collision lower bound). *With $m_C = \sum_{g \in C} c_a(g)$,*

$$\boxed{T_a \geq \mathcal{L}_{a,B} := \sum_{C \in G_a/B} \mathbf{b}_s(m_C) - \mathcal{E}_a \geq L_{a,B}.} \quad (20)$$

Equality in the first inequality holds exactly when the antipodal-pair totals within every B -coset differ by at most one. Moreover

$$\boxed{T_a \geq 4 \max\{0, \lceil \mathcal{L}_{a,B}/4 \rceil\}.} \quad (21)$$

Proof. Put $z_{C,j} = c_a(g_j) + c_a(-g_j)$ for the s pairs in a coset. Expansion gives

$$T_a + \mathcal{E}_a = \sum_C \sum_{j=1}^s z_{C,j}^2.$$

For nonnegative integers of sum m_C , transferring one unit from a coordinate at least two larger than another strictly decreases the sum of squares. It therefore has minimum (19), with exactly the stated equality locus. Real Cauchy-Schwarz gives $\mathbf{b}_s(m_C) \geq m_C^2/s$, proving the second inequality. Lemma 2.1 proves (21). $\square$

The retained spectral interpretation is exact. Let $Q_B c$ be the vector constant at $m_C/|B|$ on each coset, and $J_- c(g) = c(-g)$. In the explicitly specified Euclidean residue-coordinate metric, Q_B is an orthogonal projection and $J_- Q_B = Q_B$. Therefore

$$T_a = \|Q_B c_a\|^2 + \langle c_a - Q_B c_a, J_-(c_a - Q_B c_a) \rangle,$$

which yields (17) without assuming that the support is saturated. Fourier expansion of Q_B retains *all* characters trivial on B ; they are even. For $B = G_a$ this is the principal estimate. For $B = \{1, -1\}$ it is the exact answer, so that terminal choice is not presented as a reduced forcing theorem.

The bound uses coset counts and equal-residue collisions, not negative-target membership. When it is positive, enumerate the original residue buckets until one bucket and its negative are both nonempty, then apply (7)–(8) and (2)–(4). There are exactly $2\tau_{\text{div}}(a)$ original words, so this finite return terminates under the proved positive bound. This is a terminating return from a *verified inequality*, not a proof that the inequality always holds.

4.1. Original fibres and metrics. The map from the free integer module on original divisors to residue coefficients sums inside each actual residue fibre. Its kernel is generated by $e_d - e_{d_*}$, with d_* the least original divisor in that fibre. Retaining the sum and all nonselected coordinates reconstructs the selected coordinate by subtraction. The same construction applies to the residue-to-coset map. Thus neither projection is claimed injective. In the word-counting metric, the minimum-norm section of residue data y_g assigns $y_g/c_a(g)$ to each word in that nonempty fibre and has squared norm $\sum y_g^2/c_a(g)$. This is not the unweighted residue metric used to evaluate the explicitly defined collision energy. In particular coset summation is not declared isometric.

A positive integer coarse coefficient that reduces to zero modulo two remains supported with its original fibre. An empty fibre is absent. The proof of (20) uses the integer counts before such reduction. It never infers a fine target from coarse membership alone; the retained collision term supplies the bound.

4.2. The complement-aware refinement. There is further original information which can be retained without inspecting the target. Put $P = pa \bmod R_a = (2a)^2 \bmod R_a$. Divisor complementation gives

$$c_a(g) = c_a(P/g). \quad (22)$$

If $g^2 = P$, the coefficient $c_a(g)$ is even: the fixed residue fibre is paired by $d \mapsto pa/d$, which has no fixed original word.

Complementation pairs the B -cosets. A distinct pair C, C' has equal masses m and contributes at least $2b_s(m)$ to $T_a + \mathcal{E}_a$. For a self-complementary coset, let f be half the number of its roots of $g^2 = P$. The induced involution on its antipodal pairs has f fixed pairs and $r = (s - f)/2$ two-pair orbits. There are no extra fixed pairs from $g^2 = -P$, since -1 is a nonsquare. The coset mass is even, say $2M$. Its exact lower contribution is

$$c_{r,f}(M) = \min_{\substack{z_i, w_j \geq 0 \text{ integers} \\ \sum z_i + \sum w_j = M}} \left(2 \sum_{i=1}^r z_i^2 + 4 \sum_{j=1}^f w_j^2 \right). \quad (23)$$

This minimum is executable without an oracle: select the M smallest marginal costs in the r lists $2, 6, 10, \dots$ and f lists $4, 12, 20, \dots$, retaining the list labels. Every selected prefix is feasible. Exchange of a selected larger cost with an unselected smaller cost proves optimality. A separate dynamic program checks the finite examples. Summing these orbit bounds and subtracting $\mathcal{E}_a$ strengthens (20). It is an actual source-complement refinement, not an assumption that a general coefficient array satisfies (22).

5. FINITE COUNTEREXAMPLE TO ALL PRINCIPAL-ONLY SHELL REWEIGHTING

At the deterministically verified hard prime

$$p = 87481 \equiv 121 \pmod{840},$$

complete enumeration of all 21870 first-half shells proves

$$\boxed{L_a \leq 0 \quad \text{for every } a \in I_p.} \quad (24)$$

The integer refinement (20) with $B = G_a$ is also nonpositive at every shell. There are nevertheless 34 occupied shells. Equation (24) proves that no choice of nonnegative shell weights, even chosen after factorization, makes $\sum_a w_a L_a$ positive at this prime. This is a counterexample to that proposed certificate family, not to ES.

At $a = 21878 = 2 \cdot 10939$, $R = 31$, take

$$B = \langle -1, 2 \rangle = \{1, 2, 4, 8, 15, 16, 23, 27, 29, 30\}.$$

The original mass is eight, the energy eight, and the coarse masses are $(8, 0, 0)$. Thus $L_{a,B} = 24/5$, $\mathcal{L}_{a,B} = 6$, and (21) forces $T_a \geq 8$, attained. The pair $(1, p)$ returns the original E state

$$(h, r, s, \kappa) = (21878, 1, 1, 2822),$$

$$\frac{4}{87481} = \frac{1}{21878} + \frac{1}{61739716} + \frac{1}{5401052095396}.$$

The shell is already an elementary endpoint success. Its role here is to refute the claim that the full principal estimate is necessarily a sufficient detector, not to claim a previously uncovered prime.

6. A CUBIC QUOTIENT GIVES A SHARP ORIGINAL LOWER BOUND

At the hard prime $p = 944329 \equiv 169 \pmod{840}$, every principal-only L_a is again nonpositive. Choose the original shell

$$a = 236098 = 2 \cdot 97 \cdot 1217, \quad R = 63,$$

and the chosen observation subgroup

$$B = \langle -1, 13 \rangle = \{1, 8, 13, 20, 22, 29, 34, 41, 43, 50, 55, 62\}.$$

This has index three; it is not a saturated support or assumed stabilizer. The original 16 divisors of pa have distinct fine residues, so $\mathcal{E}_a = 16$. The three coset counts are $(8, 8, 0)$. Since $s = 6$,

$$\mathfrak{b}_6(8) = 12, \quad \boxed{\mathcal{L}_{a,B} = 12 + 12 - 16 = 8 = T_a.} \quad (25)$$

In comparison, the principal-only lower bound is $-16/9$, and the real coset bound is $16/3$. If $\omega^2 + \omega + 1 = 0$, the two retained nonprincipal cubic coefficients are $8(1 + \omega)$ and its conjugate, each of squared absolute value 64. Their combined positive energy is $32/9$ after division by $\varphi(63)$. Twice that energy is exactly the improvement from $-16/9$ to $16/3$. The integer balancing supplies the remaining sharpening to eight. Both occupied coarse counts reduce to zero modulo two; their original words and the integer argument do not disappear.

The source pair $(1, 1149248393) = (1, 944329 \cdot 1217)$ returns

$$(u, h, r, s, \kappa) = (287331266, 194, 1217, 1, 18242038),$$

$$\frac{4}{944329} = \frac{1}{236098} + \frac{1}{3538955372} + \frac{1}{4067138774169717196}.$$

At the same prime, the occupied shell $a = 236094$, $R = 47$ has $T_a = 60$ and principal lower bound $-732/23$. Its unit group has order 46. There is no proper subgroup containing -1 with order greater than two. Thus this shell has no nonterminal subgroup observation of the present kind. Taking the order-two subgroup computes T_a exactly and is not a new reduced certificate. This negative control prevents universal success of the quotient hierarchy from being asserted without proof.

6.1. A proper selected spectrum resolves the prime-order quotient. The absence of an intermediate subgroup at $R = 47$ does not preclude a smaller spectral observation. Keep the same original shell

$$p = 944329, \quad a = 236094 = 2 \cdot 3 \cdot 19^2 \cdot 109, \quad R = 47.$$

The element 5 generates its unit group of order 46. Let $\zeta = e^{2\pi i/23}$ and $\chi(5) = \zeta$, so $\chi(-1) = 1$. The original 48 divisors give, in these even-character coordinates,

$$C(X) = 2 + X + X^{11} + 2X^{12} + 3X^{13} + 4X^{14} + 4X^{15} + 5X^{16} + 5X^{17} + 5X^{18} + 4X^{19} + 4X^{20} + 4X^{21} + 3X^{22}.$$

For the original norm polynomial $Q(X) = C(X)C(X^{-1}) \pmod{X^{23} - 1}$,

$$Q_0 = 192, \quad (Q_1, \dots, Q_{11}) = (187, 176, 159, 138, 116, 92, 70, 50, 33, 21, 14), \quad Q_{23-j} = Q_j.$$

Exact rational intervals give

$$\boxed{1009 < Q(\zeta) < 1010, \quad 24 < Q(\zeta^2) < 25.} \quad (25a)$$

For completeness the interval certificate uses $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$. The rational tangent formula makes $\tan(4 \arctan(1/5) - \arctan(1/239)) = 1$, and that angle is in $(0, 1)$; $\pi/2 > 1$ follows from $\arctan 1 > 1/2$, so the identity has the stated branch. Use 18 alternating arctangent terms with the next-term remainder. For $0 \leq x \leq \pi$, use 18 cosine terms and the next-term absolute remainder; all tail terms are decreasing. Uncertainty in π contributes at most the argument uncertainty, since $|\cos x - \cos y| \leq |x - y|$. Evaluating $Q_0 + 2 \sum_{j=1}^{11} Q_j \cos(2\pi m j/23)$ with reflection into $[0, \pi]$ proves (25a) by rational arithmetic. A second checker uses $\pi = 4(\arctan(1/2) + \arctan(1/3))$, 50 arctangent terms and 21 cosine terms, and obtains the same strict integer enclosures. No floating-point value is a proof input. Both polynomial and intervals remain in the JSON certificate.

The collision energy is 132 and the principal squared sum is 2304. Retaining only $\{1, \chi, \bar{\chi}\}$ in (18a) gives

$$T_a \geq \frac{2Q(\zeta) - 732}{23} > \frac{1286}{23} > 52, \quad \text{hence } T_a \geq 56.$$

Adding $\chi^2, \bar{\chi}^2$ gives

$$\boxed{T_a > \frac{2(1009 + 24) - 732}{23} = 58, \quad T_a \geq 60,} \quad (25b)$$

using the original fourfold action. Direct enumeration, performed separately, gives $T_a = 60$. Thus five of the 23 even-character modes certify the full attained lower bound. Unlike retaining every even mode, this is a proper observation. The original pair (2, 327) returns the M state

$$(u, h, r, s, \lambda) = (1444, 361, 2, 327, 7),$$

$$\frac{4}{944329} = \frac{1}{236094} + \frac{1}{780326438241} + \frac{1}{4772638766}.$$

The norm polynomial is calculated from the original words; its identity $Q_0 = \mathcal{E}_a + T_a$ is a verification check, not an assumed target input to the interval inequality. The retained observation consists of the selected Fourier values and collision energy; the remaining Fourier coordinates form its explicit kernel complement. Orthogonal Fourier projection onto a conjugate-stable set of even modes commutes with J_- , which is the identity on that image. Keeping all omitted coordinates gives its inverse. No coset saturation, change of coefficient prime, or positivity of the omitted signed part is postulated.

7. FINITE COMPARISON AND THE REMAINING UNIVERSAL INEQUALITY

The complete declared hard universe through 2,000,000 has 4519 primes. The principal-only condition $L_a > 0$ at some original shell holds at 4515. Its exact misses are

$$87481, \quad 196561, \quad 944329, \quad 1915201. \quad (26)$$

Completeness of the finite test does not depend on an arbitrary residual cutoff: $\mathcal{E}_a \geq 2\tau_{\text{div}}(a)$ makes $L_a > 0$ require $\varphi(R) < 4\tau_{\text{div}}(a)$. The exact maximum of $\tau_{\text{div}}(a)$ over $1 \leq a \leq 999999$ is 240. All 311 possible residuals with $R \equiv 3 \pmod{4}$, $R < 2,000,000$ and $\varphi(R) < 960$ are tested; their largest is 1995. A separate implementation additionally enumerates *every* first-half shell of all four exceptions, not only this reduced candidate set.

Chosen quotient certificates for the four primes have $(a, R, |B|, \mathcal{L}_{a,B}, T_a)$ respectively

$$(21878, 31, 10, 6, 8), \quad (49159, 75, 20, 4, 16), \\ (236098, 63, 12, 8, 8), \quad (478836, 143, 4, 86, 144).$$

The last complement-aware lower bound is 88. At the second prime, even the integer principal observation at $a = 49147$, $R = 27$ gives lower bound two where the real bound is zero. The certificates retain this simpler additional repair. The basic integer scalar test therefore covers 4516 of the 4519 primes; the displayed nonprincipal certificates cover the remaining three. These are comparisons of sufficient tests, not newly solved or previously unverified primes.

The complete small replay checks all 73798 first-half shells at the 211 primes $p \equiv 1 \pmod{4}$, $p \leq 3000$. It retains 1877004 divisor vectors, 5700 E states, 5268 oriented M states, and 70596 target pairs. The independent implementation uses literal pair comparisons and trial enumeration of square divisors on that whole range. Finite computations do not establish a positive lower bound at all primes. Theorem 3.2 and Corollary 3.3 have general written proofs and do not extrapolate from this scan.

The exact surviving sufficient quantity is

$$\sum_{a \in I_p} 4 \max\{0, \lceil \mathcal{L}_{a,B_a}/4 \rceil\}, \quad (27)$$

or its complement-aware refinement. It is bounded above by $\sum_a T_a$ and returns original witnesses when positive. No proof that (27) is positive for some nonterminal choices at every prescribed prime is supplied. Likewise no universal upper bound on the remaining signed Fourier contribution has been proved. The source's requested universal closure is therefore unresolved.

The next estimate must force enough of the actual even-character energy without assuming a favourable spectrum, or exploit original off-diagonal correlations before absolute values, or supply a different arithmetic witness. The preceding factor-supply character is annihilated in the combined observation [2, §4, “The source character is annihilated”]; repeating that fact cannot provide the missing positive term. Averaged ES counts [1, Theorem 1.1 and Remark 1.2] are not converted into a pointwise lower bound. There is no Lean build, independent mathematical review, or historical priority assertion in this tranche.

REFERENCES

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- [3] The Clankers. (2026, September 9–12). *Boolean product formulation* [Exported research conversation]. Original supplied transcript, lines 7859–7884 (ordered divisor-pair count and character expression), lines 8140–8196 (the unproved full-shell inequality), and lines 8419–8435 (an earlier one-shell cancellation example). These are history and antecedent locators, not a claim that every assertion in the conversation is valid.
- [4] Farah, C. (2026, April). *Semi-constructive approach to the Erdős–Straus conjecture* [Author-uploaded preprint]. <https://www.researchgate.net/publication/403742713>. Locator compared: Lemmas 8–10, pp. 11–12, the reciprocal-totient divisor identity and odd-progression expansion. The present explicit upper bound is proved independently. No probabilistic conclusion from this preprint is imported.